Muhammad Mahasin Irfan

Health AI and Human–AI Evaluation

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Research Summary

I am an undergraduate researcher at LUMS interested in how to evaluate AI systems when people use them to make decisions. I am helping run a registered randomized study of LLM-assisted medical triage and diagnosis, from interface design and the pre-analysis plan through participant recruitment. My other work develops evaluation methods for learning systems and studies technology use in Pakistan.

Research Interests

Health AI; human–AI decision making; trustworthy evaluation; experimental design; accessible technology.

Education

Lahore University of Management Sciences (LUMS) Lahore, Pakistan
B.S. in Computer Science; CGPA: 3.85/4.00 *September 2023 – May 2027 (expected)*

- Dean's Honor List, 2023–24, 2024–25, and 2025–26.
- Selected coursework: Operating Systems; Computer Systems; Networks and Systems; Data Structures; Foundations of Artificial Intelligence and Machine Learning.

Publications

Accepted

- **M. M. Irfan**, M. S. Hassan, and F. Zaffar. "Decomposing Error in Continual World Models." NeurIPS 2026 Workshop on Continual World Models, Idea Track. **Accepted** First author. [Code]

Submitted Papers

- Humza Ikram, **Muhammad Mahasin Irfan**, Muhammad Abdullah, Muneeb Pervez, Hashim Sharif, Akash Kothari, Yifan Zhao, and Muhammad Fareed Zaffar. Manuscript on characterizing memory-resident acceleration for edge inference. Submitted to **MLSys 2027**. **Second author.**

In preparation

- **M. M. Irfan*** et al. Security and Privacy Intermediaries in Mobile-Phone Shops (working title). Joint first author (*equal contribution).

Research Experience

Evaluating LLM Assistance for Medical Triage and Diagnosis Ongoing
Researcher, LUMS; ClinicalTrials.gov-registered randomized study *Lahore, Pakistan*
Supervised by Dr. Ihsan Ayyub Qazi (LUMS)

- Co-designed a randomized study of how LLM assistance affects medical triage and diagnosis decisions in scenario-based tasks with university staff; target enrollment is 230.
- Wrote the pre-analysis plan, selected covariates, and specified mixed-effects models and statistical tests before outcome analysis.
- Designed and built the participant-facing platform, using controlled information presentation informed by human-centered design and dark-pattern research to study attention and framing effects.
- Lead participant recruitment and data collection. The study is ongoing, and results will be reported after the planned analysis.

Decomposing Error in Continual World Models July 2026 – Present
First-Author Researcher, LUMS; NeurIPS 2026 Workshop on Continual World Models (Idea Track) *Lahore, Pakistan*
Supervised by Dr. Fareed Zaffar (LUMS)

- Derived an exact decomposition of post-update prediction error into inherited error and update mismatch, showing why endpoint accuracy alone may mis-rank continual model updates.
- Demonstrated opposite endpoint/mismatch rankings in 544/544 controlled one-step trials; the reversal persisted in 93.8% of trials at horizon 5 and 66.2% at horizon 12.
- Reproduced the phenomenon across 500 learned-model runs on a nonlinear damped-pendulum simulator; reversal frequency fell from 95% to 6% as inherited error decreased from 0.038 to 0.004.
- Developed the mathematical formulation, implemented the experiments, analyzed the results, and wrote the first-author manuscript. [Code]

Security and Privacy Intermediaries in Mobile-Phone Shops June 2026 – Present

Joint-First-Author Researcher, LUMS/CIRCLE; collaboration with CISPA *Lahore, Pakistan*

Supervised by Dr. Mobin Javed (LUMS), Dr. Katharina Krombholz (CISPA), and Dr. Lea Grober (formerly CISPA, ICSI Berkeley)

- Co-designed an IRB-approved qualitative study of mobile-phone shopkeepers as security and privacy intermediaries for low-income communities in Pakistan.
- Conducted 15 semi-structured Urdu interviews in Lahore on account creation and recovery, device migration, authentication, privacy advice, and fraud; each interview lasted approximately 30 minutes.
- Transcribed, translated, and coded interviews and conducted collaborative reflexive thematic analysis.

PIM Sizing for Edge ML Inference November 2025 – Present

Second-author researcher, LUMS; MLSys 2027 submission *Lahore, Pakistan*

Supervised by Dr. Fareed Zaffar (LUMS), Dr. Hashim Sharif (PhD, UIUC; AMD Research), and Humza Ikram (CMU PhD student)

- Helped develop a tool that compares PIM channel counts, added versus converted channels, chip designs, command issuing, and GPU/PIM schedules for latency and energy per inference.
- Built repeatable measurement and experiment pipelines across an AMD Ryzen AI laptop and an NVIDIA Jetson Orin Nano, using Triton and TensorRT kernels, scripts, and Docker environments.
- Extended the Newton PIM model in Ramulator using C++ to study bank groups, row sizes, and activation timing; contributed to LPDDR5/LPDDR5X PIM simulation. Earlier prototype configurations improved simulated throughput by 1.3× and 1.5× over the original model. [Code]
- Characterized LLM generation, vision, CNN, and robot-control workloads on two edge systems. The submitted study projects up to 13.6× faster LLM generation with simulated PIM, while vision workloads gain little.
- Earlier prototyped a prediction-guided split search that evaluated up to 7–8× fewer candidates while reaching within 1% of the measured best latency in selected cases. Nonmonotonic TensorRT curves led us to use measured timings in the submitted study.

Quantizing Diffusion Models June 2025 – August 2025

AI Optimization Intern, 10x Engineers *Lahore, Pakistan*

Supervised by Dr. Saad Bin Nasir (10xEngineers; formerly Qualcomm) and Dr. Khurram Usman Mazher (formerly Qualcomm)

- Extended QuantX for simulated post-training quantization of diffusion pipelines, adding component-aware traversal and configurable linear/convolutional weight and activation granularities for Stable Diffusion 1.5 and SDXL, with exploratory SD3.5 support. [Code]
- Built diffusion-trajectory calibration and activation-statistics infrastructure and implemented SmoothQuant-style equivalent scaling; studied PTQ4DM, Q-Diffusion, ViDiT-Q, and SVDQuant to guide the design.
- Diagnosed catastrophic W4 degradation: excluding the convolution path reduced LPIPS from 0.5713 to 0.3380, while subsequent smoothing improved it to 0.3357 under the recorded evaluation protocol.
- Benchmarked TorchAO, Quanto, and QuantX and identified sample-, dataset-, compilation-, and random-state-related evaluation pitfalls; documented the work in a public technical report.

Academic Service

Reviewer, NeurIPS 2026 Workshop on Continual World Models September 2026

Selected by the workshop Technical Program Committee

- Reviewed a workshop submission, assessing what the paper demonstrated and where it needed more evidence, and submitted a written review and recommendation to the committee.

Teaching Experience

Teaching Assistant, Computer Systems and Assembly Language (CS225) Fall 2026

LUMS *approx. 270 students, two sections*

- Develop quizzes, assignments, and a new systems laboratory; grade examinations and conduct tutorials.

Teaching Assistant, Foundations of Data Science Spring 2026

LUMS *approx. 200 students*

- Designed introductory Python and data-visualization assignments, automated tests, and Jupyter/Otter-Grader workflows; held office hours, supported course projects, graded work, and invigilated examinations.

Teaching Assistant, Introduction to Programming (CS100) Spring 2026
LUMS *approx. 180 students*

- Created assignments, held programming labs and a tutorial, and provided student-facing office hours.

Honors and Awards

- **Dean's Honor List**, LUMS: selected in every completed academic year (2023–24, 2024–25, 2025–26).
- **National Mathematics Talent Contest**: top 50 chosen from 4000+ candidates.
- **Purple Comet! Math Meet**: first-place team in Pakistan.
- **Southeast Asian Mathematical Olympiad (SEAMO)**: Gold Medal; third nationally in Pakistan.
- **Full A-Level Scholarship**: 100% tuition scholarship.

Technical Skills

Programming:	C++, Python, C; Bash; Git; Linux; Docker; Jupyter; LaTeX.
ML systems:	PyTorch; TVM; TensorRT; model quantization; diffusion models; performance evaluation and autotuning.
Architecture:	Ramulator/Newton PIM; DRAM architecture; heterogeneous GPU/PIM execution; memory-system simulation.
Research methods:	Empirical evaluation and benchmarking; semi-structured interviews; reflexive thematic analysis.
Languages:	English, Urdu, Punjabi.